

# Aravindh B

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## SUMMARY

**Software Development Engineer** with 7+ years of experience across **backend engineering**, **data pipelines**, and **cloud-native solutions**. Proficient in **Java (Spring Boot)**, **TypeScript**, and **AWS (CDK, Lambda, App Mesh)**. Delivered scalable **microservices**, **ETL frameworks**, and real-time streaming solutions with **Kafka and Informatica**. Adept in **CI/CD**, **API development**, and observability for distributed systems.

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## PROFESSIONAL EXPERIENCE

### Software Development Engineer | Amazon Services LLC | Seattle, USA

May 2022 – Jan 2025

- Engineered a centralized monitoring library using **AWS CDK** and **CloudWatch**, auto-generating 25+ dashboards with real-time anomaly detection across ECS and Lambda services.
- Formulated a reusable CI/CD pipeline framework with approval stages, bake times, and auto-scaling using **TypeScript** and **AWS Pipelines**, accelerating deployment setup by **40%**.
- Refactored the Mars platform into a distributed microservices architecture using **AWS App Mesh**, improving routing latency and service resilience during scale events.
- Decoupled legacy third-party integrations (LMS, media, token services) from Horizonte using **Java** and **Spring MVC**, increasing platform autonomy and code maintainability.
- Defined stateless service contracts and fallback handlers to ensure backward compatibility for the MaRS rendering system, reducing failure propagation across downstream APIs.
- Instrumented third-party service metrics collection and integrated them with **CloudWatch**, enabling consistent reporting and alerting for 10+ critical microservices.
- Constructed UI enhancements on Alexa Echo Dot using **jQuery**, **AUI**, and **JavaScript**, introducing tab-based prompts that improved accessibility for voice command users by **25%**.
- Established IAM policies, PrivateLink endpoints, and multi-account security protocols for widget-based services, hardening system access across 5 VPCs.
- Streamlined deployment and observability of widget services across accounts by aligning **API Gateway**, **Lambda**, and **CDK constructs**, reducing onboarding time by **35%**.

### Software Engineer | Tata Consultancy Service | Remote, USA

Nov 2021 – May 2022

- Implemented **10+ backend endpoints** using **Spring Boot**, **Spring MVC**, and **Spring Data JPA**, enabling smooth data flow across services and reducing backend latency by **20%**.
- Developed secure API logic using **Spring Security** and **JWT**, managing role-based access control and decreasing unauthorized access incidents during UAT by **35%**.
- Integrated both **REST** and **GraphQL APIs** into backend services to support flexible querying and efficient data retrieval for high-traffic client apps.
- Built responsive web modules in **Angular** using **HttpClientModule**, **FormsModule**, and **RouterModule**, enhancing the UI experience across 3 external-facing applications.
- Connected **React Native** frontends with backend APIs using **Axios**, **fetch**, and **GraphQL**, enabling real-time synchronization across both **Android** and **iOS** platforms.

### Sr. Tech Associate, Data Engineer | Bank of America | Hyderabad, India

Dec 2017 – Dec 2019

- Designed and maintained ETL pipelines using **Informatica PowerCenter** and **SSIS**, extracting and transforming data from **Oracle**, **SQL Server**, and **Access** for 10+ enterprise data marts.
- Authored 40+ mapping workflows using transformations like **Lookup**, **Aggregator**, **Joiner**, and **Update Strategy**, enabling real-time data migration from structured and semi-structured sources.
- Converted 120+ unstructured Excel files to CSV using **Python scripts**, and automated ingestion into ETL pipelines, reducing manual prep time by **60%**.
- Scripted Unix-based file movement routines to pull source data from SharePoint and archive directories, scheduled through **Autosys** for daily batch runs across 4 environments.
- Developed complex **PL/SQL stored procedures**, triggers, and functions to support rule-based transformations, optimizing performance of critical jobs by **35%**.
- Integrated **Kafka** to stream live transaction data into **HDFS**, supporting downstream analytics with <5-second lag across high-volume ingestion channels.
- Constructed reusable **Mapplets** and transformation templates in **Informatica**, improving consistency and reducing development time by 25% for new pipelines.
- Partnered with business analysts and QA teams to validate ETL logic and coordinate release schedules, resolving over 50 production issues across 18 months.

### Sr. Tech Associate, Software Developer | Bank of America | Hyderabad, India

Jul 2017 – Dec 2017

- Assembled backend microservices using **Spring Boot** and **Spring MVC** to support a Learning Management System, enabling seamless communication between 3+ internal applications and external consumers.
- Coded secure login and registration APIs using **Java 8** and **Spring Security**, improving authentication workflows and reducing onboarding errors by **40%** during integration testing.

- Constructed responsive UI screens using **React.js**, **jQuery**, **HTML5**, and **CSS3**, enhancing training module navigation and increasing user engagement for 200+ internal users.

**Junior Software Developer | KPMG | Hyderabad, India**

*May 2016 – Jun 2017*

- Developed 5 internal scripts in **Python** to automate data reconciliation across financial audit reports, reducing manual validation time by **30%**.
- Wrote complex **SQL** queries and used **pandas** for data transformation on Oracle and DB2 sources, improving processing speed during client reporting cycles by **25%**.
- Tested multiple RESTful endpoints using **Postman** and **Python’s requests module**, validating data exchange between compliance apps and third-party platforms.
- Built static UI templates with **HTML5**, **CSS3**, and basic **Flask routing**, which were used in prototypes for internal tax analysis tools.
- Created 6 reusable Python modules for exception handling, string formatting, and time-based triggers to simplify logic across multiple reporting tasks.
- Tracked code versions and collaborated using **Git** and **JIRA**, resolving merge issues and supporting sprint updates alongside senior Python developers.

TECHNICAL SKILLS

- **Programming Languages:** Java (8+), TypeScript, JavaScript (ES6+), Python, R, Shell Scripting, C, C++
- **Frameworks & Libraries:** Spring Boot, Spring MVC, Microservices, Hibernate, Node.js, Express.js, React.js, Angular, React Native, jQuery
- **Web Technologies & APIs:** REST, SOAP, GraphQL, AJAX, HTML5, CSS3, JSON, XML
- **Cloud Platforms:** AWS (CDK, EC2, ECS, Lambda, RDS, DynamoDB, VPC, Route53, ELB, App Mesh, API Gateway, Transit Gateway, PrivateLink, IAM, SNS, SQS, CloudWatch, Pipelines)
- **DevOps & CI/CD Tools:** Jenkins, Docker, Ansible, GitHub, GitFarm, Bitbucket, IntelliJ, VS Code, Eclipse
- **Databases:** Oracle, MySQL, SQL Server, DB2, MongoDB, DynamoDB
- **ETL & Data Engineering Tools:** Informatica PowerCenter, SSIS, Autosys, Toad for Oracle, SQLLoader, Trifacta Wrangler, Kafka, HDFS
- **Data Processing & Scripting:** SQL, PL/SQL, Unix Shell Scripting, Python (for automation & ETL), Java-based ETL pipelines
- **Authentication & Security:** Spring Security, OAuth 2.0, JWT, IAM Policies
- **Project & Collaboration Tools:** JIRA, Confluence
- **Version Control & Build Tools:** Git, Maven, CI/CD Pipelines

PROJECTS

Distributed File System Architecture with RPyC

- Built a distributed file system simulating HDFS with one master and three data nodes using **RPyC** for remote procedure calls, improving fault tolerance and replication control by over **80%**.
- Enabled concurrent file access and logging through **multithreading**, supporting **simultaneous client interactions** with no data conflict during stress testing.

Market Basket Analysis Using MapReduce

- Mined **10,000+** shopping transactions using the **Apriori algorithm** in **MapReduce**, identifying frequent itemsets and boosting pattern detection speed by **30%**.
- Created a front-end interface in **Node.js** to display association rules and user behavior insights, with real-time data queries handled via **MongoDB**.

EDUCATION

**Westcliff University | Irvine, CA**

*Sep 2024 – Sep 2027*

Doctorate in Business Administration

**Arkansas State University | Jonesboro, AR**

*Jan 2020 – Jul 2021*

Master’s in Computer Science and Engineering

**Chaitanya Bharathi Institute of Technology | Hyderabad, India**

*Aug 2013 – May 2017*

Bachelor’s in Computer Science and Engineering